

Protein Structure Prediction- CodeAlpha Internship

Bioinformatics

Protein Selected: Human Insulin

UniProt ID: P01308

Tool: PyMOL

Method Used: AlphaFold DB

1. Structure Overview

The predicted 3D structure of human insulin shows a compact, globular conformation composed of two polypeptide chains (A and B chains). Disulfide bonds stabilize the structure.

2. Secondary Structure

The protein contains:

- Alpha helices in both A and B chains
- Short loops connecting structural regions
- No extensive beta-sheet regions

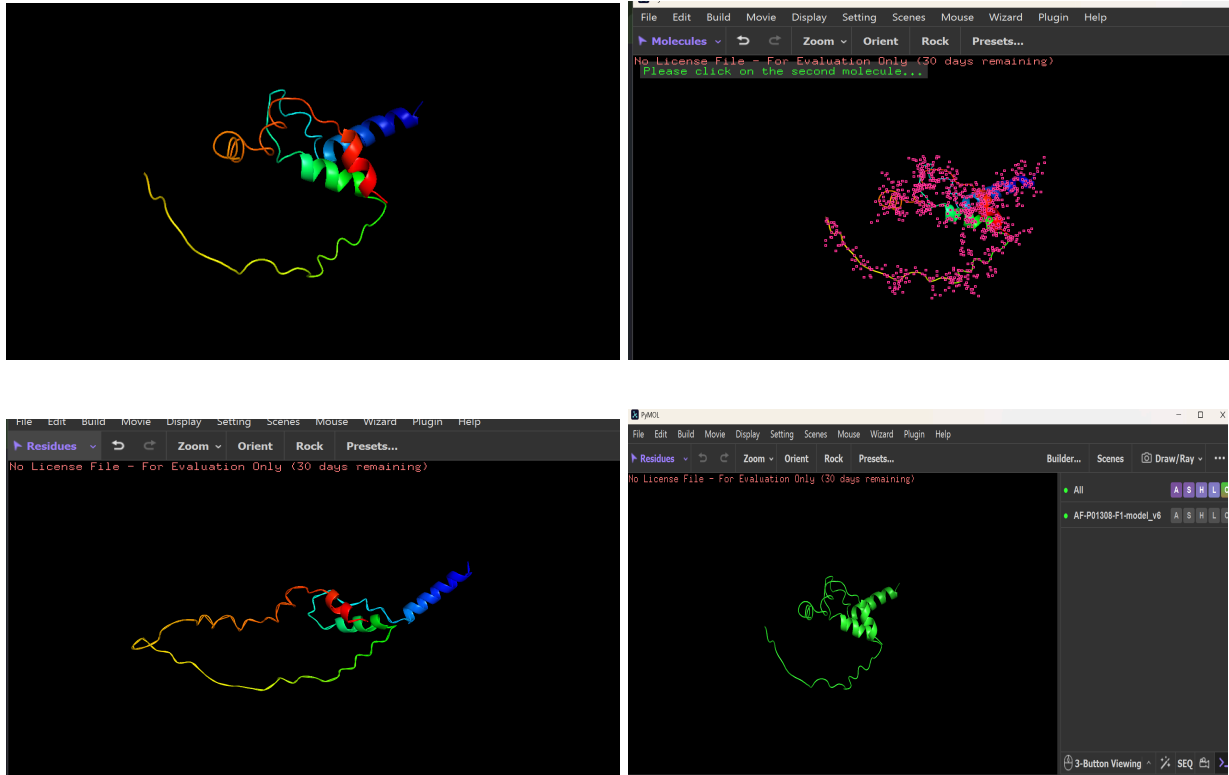
3. Key Structural Features

- Presence of **disulfide bonds** between cysteine residues, which stabilize the structure
- Hydrophobic core that maintains folding
- Surface-exposed residues likely involved in receptor binding

4. Functional Insights

This is a globular protein. The three helical segments fold tightly together into a compact, spherical bundle. The long, floppy orange/yellow tail trailing off to the left is the flexible connecting peptide region of the precursor form (proinsulin).

5. Visualization



Conclusion

The predicted structure highlights conserved structural elements essential for insulin function, especially disulfide bonds and alpha-helical regions.